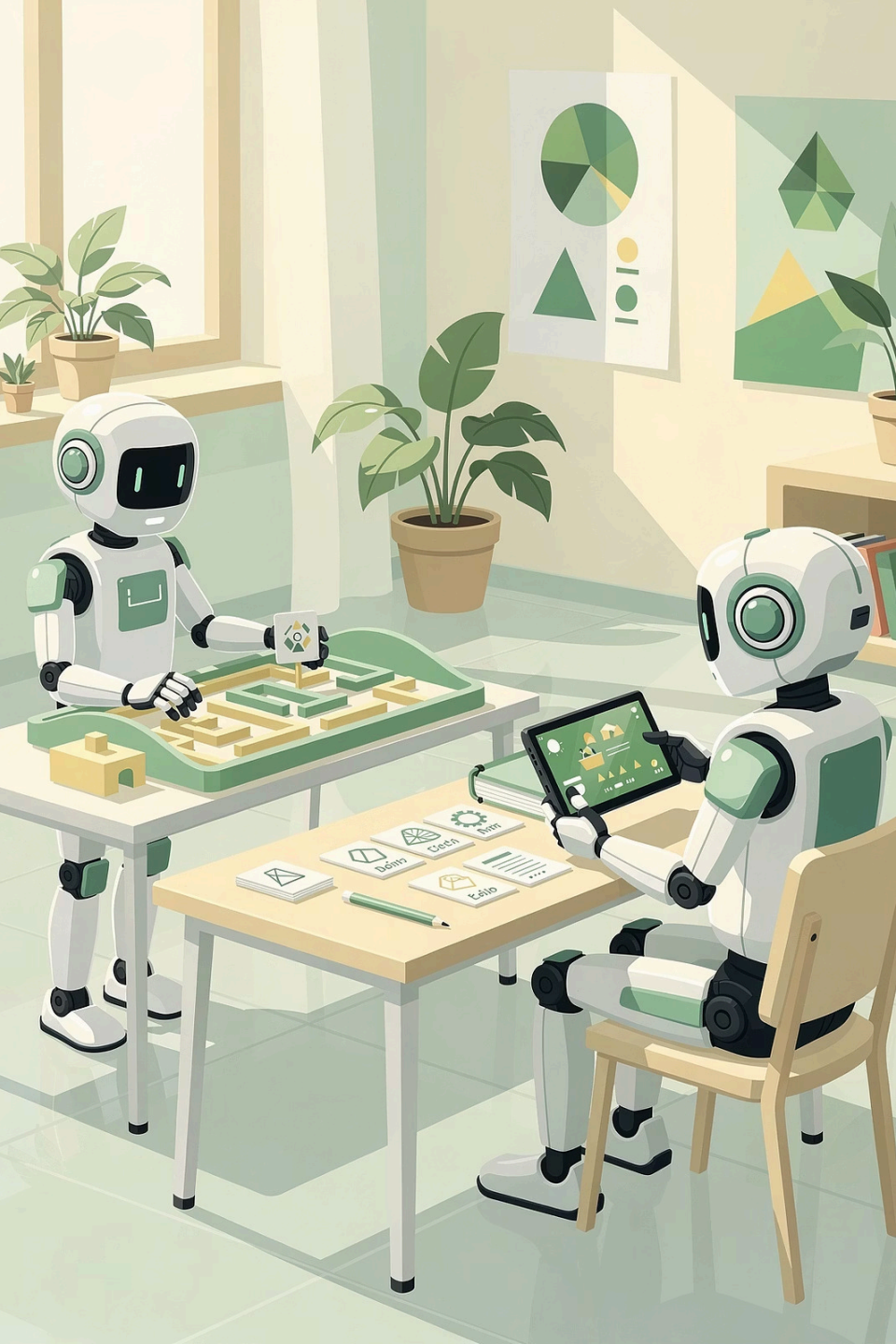




Session 4: Types of Machine Learning

Last session you saw **HOW** machines learn (patterns from examples).

In this session you'll find out the **different ways** they learn - there's more than one!

❏ Spoiler: it's a lot like how **YOU** learn different things in different ways.

Objectives

By the end of this session, you will be able to:

01

Know the 3 Main Ways

Machines can learn in three distinct styles - and each one is built for different problems.

02

Explain Supervised Learning

Learning **with an answer key** - the machine gets examples that already have the right answer.

03

Explain Unsupervised Learning

Finding patterns **alone** - no labels, no answers, just exploration.

04

Explain Reinforcement Learning

Learning **like a video game** - try things, get rewards, get better.

05

Match Apps to Learning Types

Look at real apps and identify which type of learning powers them.

Recap & Homework

Welcome back! Let's see what you found.

ML Hunt - Show Me Your 3 ML Systems

- Which apps did you pick?
- What patterns do you think they learned?

Quick Check from Last Time

- What's Machine Learning in your own words?
- Rules vs. Learning - what's the difference?



Warm-Up: How Do YOU Learn?

Think about all the different ways **YOU** learn things:



Flashcards

Sometimes you learn with **flashcards** that have answers on the back - you check if you got it right!



Exploring

Sometimes you learn by **exploring** and figuring things out yourself - no one tells you the answer!



Trying Things

Sometimes you learn by **trying things** and seeing what works - trial, error, and discovery!

☑ Different ways of learning - for different kinds of stuff. **Guess what? Machines learn the same three ways!**

The Big Idea: 3 Ways Machines Learn

Just like you, machines have different learning styles. Here's the big picture:

- 🎓 **Supervised Learning**
Learning **WITH** a teacher or answer key
- 🔍 **Unsupervised Learning**
Learning **ALONE** by exploring
- 🎮 **Reinforcement Learning**
Learning by **trial and error** - like a video game!

Today we'll dig into all three. Let's go! 🚀



Supervised Learning

Supervised Learning = the machine learns from examples that already have the **right answer**.

We call this **labeled data** - like flashcards where the answer is on the back.

Examples in the Real World:

- 📧 Spam Filter
Trained on emails labeled "**spam**" or "**not spam**"
- 📷 Face Unlock
Trained on photos labeled "**this is YOU**"
- 🐱 Cat vs Dog
Trained on photos labeled "**cat**" or "**dog**"

📌 The key: **Every example has the correct answer attached.**



Supervised Learning Has Two Jobs

Supervised AI does one of two things - but both use the same kind of learning. They just answer different questions.

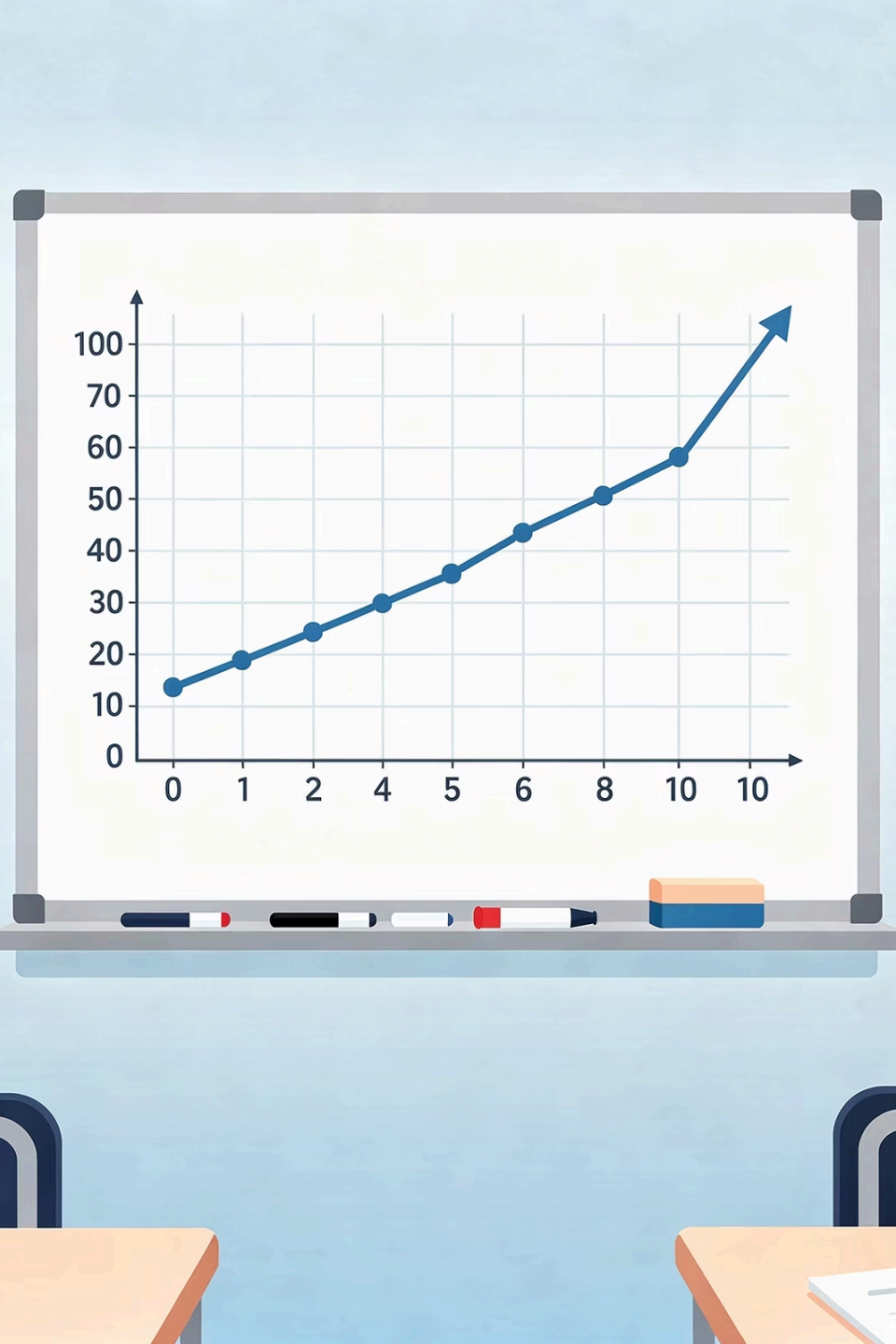
Classification

Regression

Classification - Sorting Things into Categories

- 📧 Spam or not spam?
- 🐱 Cat or dog?
- 😊 Happy face or sad face?





Regression Predicting a Number 1 2 3 4

- 🌡️ What's tomorrow's temperature?
- 🏠 How much will this house cost?
- 🚗 How long will my drive take?

Both jobs use labeled training data - the difference is just whether the output is a **category** or a **number**.



Unsupervised Learning

Unsupervised Learning = the machine gets examples but **NO answers attached**. No labels. It has to find patterns on its own.

Think of it like **organizing a really messy closet** - nobody tells you how, you just notice what goes together.

Example: 1,000 Unlabeled Photos



"These 200 look similar..."

They're all beach pictures



"These 300 look similar..."

They're all food photos



"These 500..."

All selfies



Nobody told it. It just grouped them by what looked alike.

Where You See Unsupervised Learning

Apps you use every day that do this - the AI finds groups **all by itself**. Nobody tells it the answer.



Google News

Groups news articles about the **same topic** together



Amazon

"Customers also bought..." - finds groups of people with **similar shopping habits**



Spotify Discover Weekly

Finds songs that **"go together"** by what they sound like



Google Photos

Groups all photos of the **same person** together - without you labeling them!

Reinforcement Learning

Reinforcement Learning = the machine learns by **trying things** and seeing what works - and getting "rewards" when it does the right thing.

Like playing a video game, but the "game" can be anything: driving a car, walking without falling, beating a chess champion.

Try Something → It Works!

Reward! 🎉 The AI remembers this action.

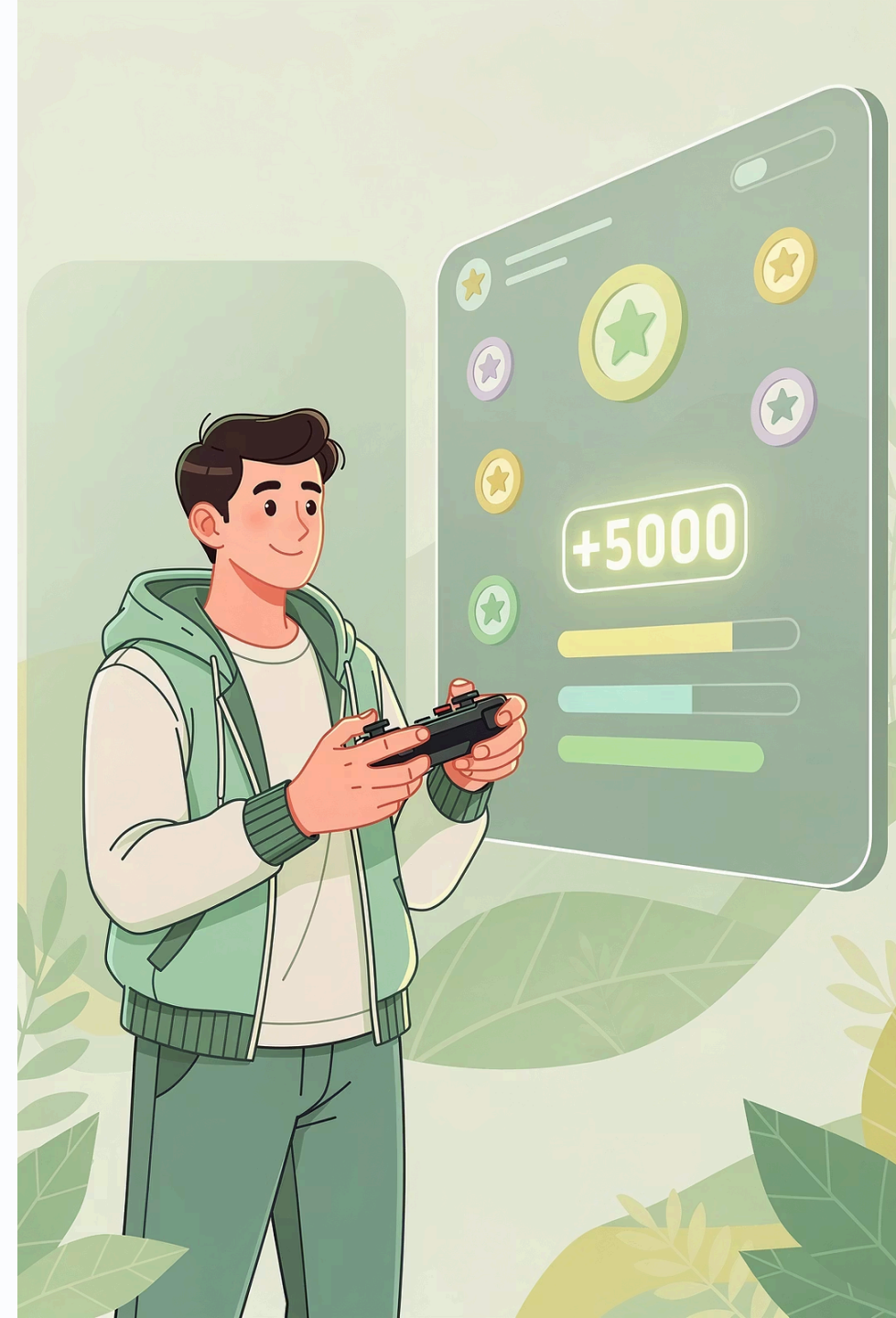
Try Something → Doesn't Work

Penalty! ❤️ The AI avoids this next time.

Over Time → Gets Better

Learns what gets the **most rewards** through millions of tries.

Over time, it learns what gets the most rewards. The AI doesn't get a rulebook - it just tries stuff.



Reinforcement Learning in Real Life

Try an Action

Receive Reward



Update Strategy

Receive Penalty

Real examples of machines learning through trial, error, and rewards:



AlphaGo

Learned to play Go by playing **millions of games against itself** - no human teacher needed.



Self-Driving Cars

Learn to stay in their lane by getting **rewards for safe driving** and penalties for mistakes.



Robot Dog

Learns to walk by **trying, falling, getting up**, and trying again - just like a toddler!



Game AI

Enemies in games learn what makes **you fail** and get progressively harder over time.

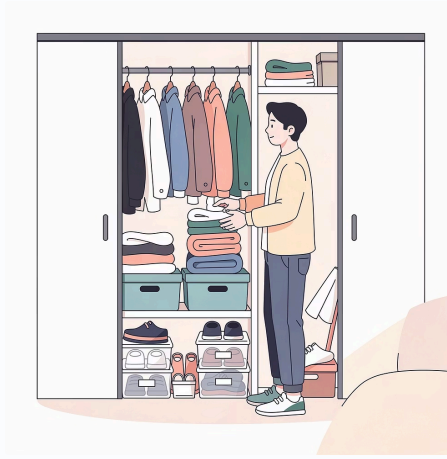
Quick Pattern: All 3 Types Side by Side

Let's see them together so the differences are crystal clear.



🎓 Supervised Learning

- Trained on labeled data
- Predicts known categories or values
- Best for: spam detection, image classification



🔍 Unsupervised Learning

- Finds hidden structure in unlabeled data
- Groups similar items together
- Best for: customer segmentation, anomaly detection



🎮 Reinforcement Learning

- Learns through rewards and penalties
- Improves strategy over many attempts
- Best for: robotics, game AI, autonomous driving

📌 **Each one is good for different kinds of problems.** The trick is knowing which to use!



Mini Activity 1: Match the Type

Which type of learning is each one? Choose: **Supervised**, **Unsupervised**, or **Reinforcement**

Scenario	Your Guess
Predicting tomorrow's weather	?
Spotify grouping similar songs together	?
An AI learning to play chess	?
Detecting if an email is spam	?
Google News grouping articles by topic	?
A robot learning to walk	?
Recognizing if a photo shows a cat	?

Think it through - look at the data and the goal for each one!

Mini Activity 1: Answers

Scenario	Answer
 Predicting tomorrow's weather	 Supervised (predicting a number)
 Spotify grouping similar songs	 Unsupervised
 AI learning to play chess	 Reinforcement
 Email spam detection	 Supervised (sorting)
 Google News grouping	 Unsupervised
 Robot learning to walk	 Reinforcement
 Cat or not cat?	 Supervised (sorting)

Need answers in advance?

→  **Supervised**

No answers, figures out groups?

→  **Unsupervised**

Tries things, learns from rewards?

→  **Reinforcement**

Think About It This Way...

Imagine you're learning to bake cookies - **three different ways**. Same goal. Three completely different paths to get there.

Supervised

Mom gives you the **exact recipe with measurements** AND shows you the perfect cookie. You copy what she does. You have the answer before you start.

Unsupervised

You look at **LOTS of cookies** and notice patterns. "Flat ones are crispy. Round ones are chewy." You group them by what you observe - no recipe needed.

Reinforcement

You try different recipes. **Yummy = WIN!**  **Burnt = LOSE.**  You remember what worked and keep improving with every batch.





Mini Activity 2: You Be the Coach

You're designing an AI. Which type should it use? For each problem, pick the right learning type.

1

Video Game AI

An AI that learns to play your favorite video game → _____

2

Photo Library Sorter

An AI that sorts your messy photo library by people → _____

3

Dog or Cat?

An AI that tells you if a photo shows a dog or a cat → _____

4

Friend Group Finder

An AI that figures out which kids in your class are friends (by who hangs out together) → _____

5

YouTube Recommender

An AI that gets better at recommending YouTube videos to you → _____

Mini Activity 2: Answers

1 Video Game AI → Reinforcement

Gets rewards for winning, penalties for losing - learns the best strategy over time.

2 Photo Sorting → Unsupervised

No labels given - just groups photos by who looks similar.

3 Dog or Cat? → Supervised

Trained on labeled examples: "this is a dog," "this is a cat."

4 Finding Friend Groups → Unsupervised

No list of "friends" given - just finds patterns in who spends time together.

5 YouTube Recommendations → + + All Three!

Groups similar viewers (Unsupervised), learns from your watch history (Supervised), and tests what keeps you watching (Reinforcement).



Quick Side-by-Side: Final Cheat Sheet

Everything you need to remember - in one table.

Feature	🎓 Supervised	🔍 Unsupervised	🎮 Reinforcement
Data Type	Has answers (labeled)	No answers (unlabeled)	No data at start
Goal	Predict or sort	Find groups	Get most rewards
Feedback	"Right or wrong?"	None - just patterns	Rewards & penalties
Like...	Flashcards	Messy closet	Video game

Hands-On: Spot the Learning Type!

Time to be a detective. Let's look at real apps together. For each app, figure out: What kind of learning? What kind of data? What's the goal?

Spotify

Groups similar listeners, learns from skips and replays

YouTube

Recommends videos based on watch history and engagement

Google Maps

Predicts travel time from millions of past trips

Face Unlock

Trained on labeled photos of your face

ChatGPT

Learned from massive text data + human feedback rewards



Let's Try Spotify Together 🎵

Analyzing **Spotify Discover Weekly** from the inside:

1

What data does Spotify have?

What songs you play, skip, replay, add to playlists...
all your listening behavior

2

What is it doing?
All three! It **groups similar listeners**, learns from your skips, tests songs and sees if you keep them...

3

What does it predict?
The **next song you'll love** - before you even know you'll love it



You just analyzed a real AI from the inside.

Real AI Uses More Than One Type!

Here's a cool secret: **most real AI uses MORE than one type at the same time.**

Example: Netflix Recommendations

Unsupervised

Finds groups of people with **similar taste** - "people like you also watched..."

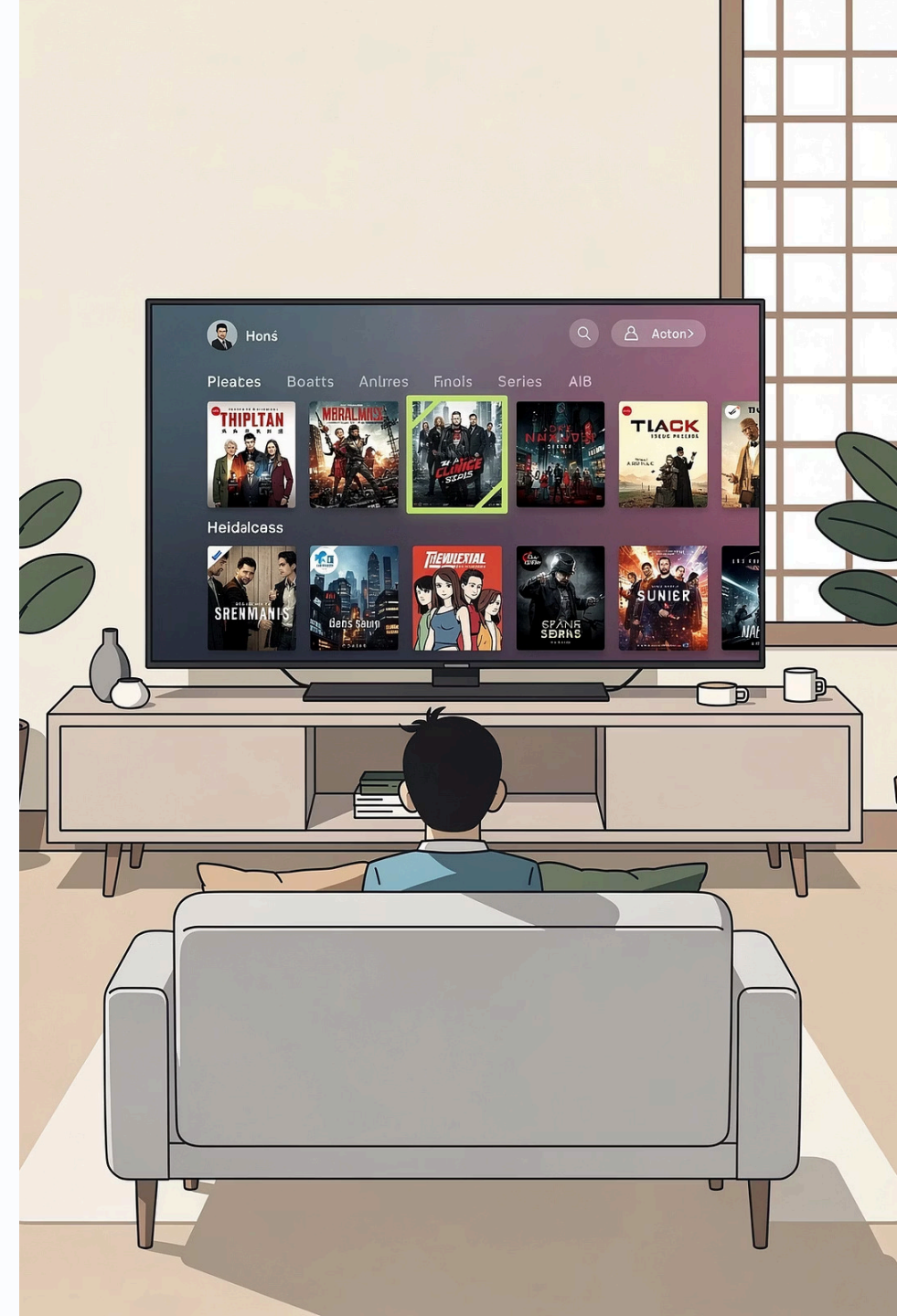
Supervised

Learns from what **YOU rate** - thumbs up, thumbs down, watch time.

Reinforcement

Tests different recommendations and **keeps what works** - optimizing for your engagement.

i One Netflix system. All three types working together. Real AI is a team effort!





What We Learned Today

You now know the **three biggest ways AI learns** - and how to spot each one by looking at the data and the goal.



Supervised Learning

Learning **with an answer key** - like studying with flashcards



Unsupervised Learning

No answers - just finds groups on its own, like sorting a messy closet



Reinforcement Learning

Trial and error with rewards - just like leveling up in a video game



Real AI Mixes Types

Real systems **combine types** depending on the job - and now you can spot which ones!

Your Homework: ML Type Hunter

Find **one example of each type** of learning. Write 1–2 sentences for each. Bring it next session.

Type	App / System	What It Does
 Supervised	_____	_____
 Unsupervised	_____	_____
 Reinforcement	_____	_____

Supervised Ideas

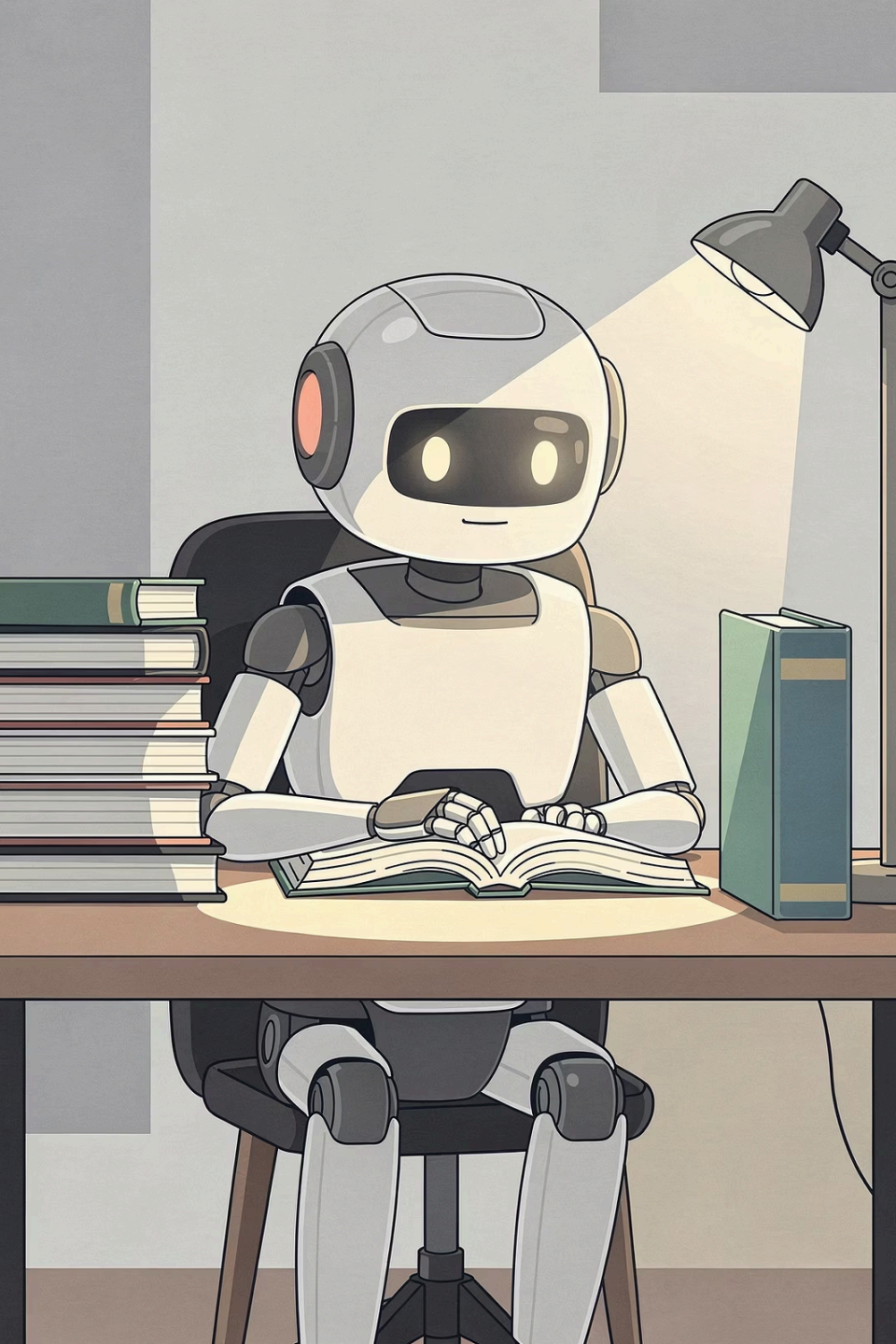
Face unlock, spam filter, translation apps

Unsupervised Ideas

Google Photos grouping faces, Amazon
"customers also bought..."

Reinforcement Ideas

Game AI, robot vacuums learning your floor
plan, AlphaGo



Next Time: Session 5 - How AI Learns from Data

Now you know the 3 ways AI learns. Next session, we go **deep on data** - the fuel that powers every AI system.



What Makes Data GOOD?

The ingredients of high-quality training data



What Makes Data BAD?

Bias, noise, and missing information



What Does AI Pay Attention To?

Features, patterns, and what the model actually sees



Garbage In = Garbage Out

Why bad data leads to bad AI - every single time



Bring your ML Type Hunter notes!  We'll use them to kick off Session 5.